UNIT-4:

Test of Hypothesis: Small Samples (<30)

If sample size is less than 30 then we can apply the test of any hypothesis are given by:

1) t distribution (means)
2) F distribution (variances)
3) χ scale distribution. (more than 2 attributes).

Degree of freedom: No. of independant variables or the no. of free variables of an equation are known as degree of freedom. ($n-1$).

Properties of t-distribution:

- It is a bell-shaped curve & it is similar to the normal distribution & is semetrical about the line ($x = \mu$) (or) ($y = 0$)
- The mean of standard normal distribution as well as t distribution is 0. but the variance of t distribution depends on the parameter μ.
- The variance of t distribution exceeds 1 but approaches 1 as n tends to ∞.
- From symmetric property of t distribution ($t_{1-\alpha} = -t_{\alpha}$).

~~from the symmetric property of t dis~~

find $t = .05$ when $\nu = 16$ from NDT

$\rightarrow$ $t_{0.05} = 1.746$.

find $-t = .01$ when $\nu = 10$ from NDT.

$\rightarrow$ $-t_{0.01} = -2.764$.

find $t = .995$ when $\nu = 7$ from NDT.

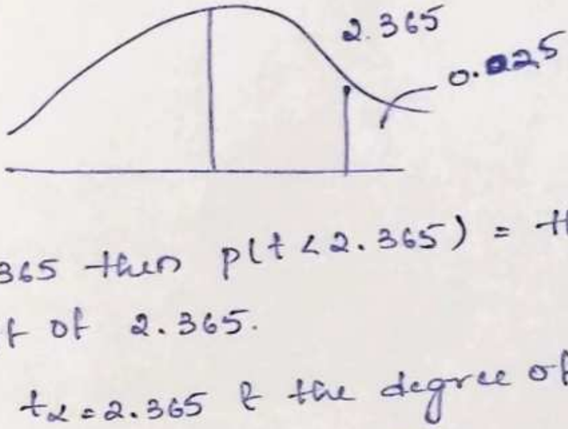
$\rightarrow$ $t_{1-\alpha} = -t_{\alpha}$

$t_{1-0.005} = -t_{005}$

$= -3.499$.

find probability of $t < 2.365$ when $\nu = 7$.

$\rightarrow$



If $t < 2.365$ then $p(t < 2.365)$ = the area to the left of 2.365.

From NDT, $t_{\alpha} = 2.365$ & the degree of freedom is 7 is then the area $\alpha = .025$

$\therefore$ $t < 2.365 = 0.975$.

find probability of $t > 1.318$ when $\nu = 24$.

$\rightarrow$ 0.10

A random sample of size 16 values from a normal population shown the mean of 53 & sum of squares of deviation from its mean is 153. Then this sample be regarded as taken from the population having the mean of 56 & also construct 95% confidence interval for the true mean.

→ Given that,

sample size $n = 16$

Sample mean, $\bar{x} = 53$

The sum of squares of deviation from its means, $\Sigma(x_i - \bar{x}) = 150$

Population mean is 56.

Null Hypothesis (H_0): The sample is drawn from the population having the mean of 56.

Alternative Hypothesis (H_1): The sample is not drawn from the population having the mean of 56

($\mu \neq \mu_0$) (TTT)

Level of significance (α): Assume $\alpha = 5\%$.

Critical value: The critical value of t ($t_{\alpha/2}$) at 5% of level of significance with 15 degrees of freedom is $\underline{\underline{2.131}}$

Test Statistics: $t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$

$s^2 = \dfrac{1}{n-1}\Sigma(x_i - \bar{x})^2$

$\dfrac{1}{15} \times 150 = 10.$

$s^2 = 10.$

$t = \dfrac{53 - 56}{\sqrt{10/16}}$

$\Rightarrow \underline{\underline{3.79}}$

Conclusion: Since the absolute value (3.79) is > than tabulated value (2.131). So reject H_0.

~~The heights of 10 males is given~~

$(1-\alpha)100\% \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$

$\bar{x} - t_{\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{s}{\sqrt{n}})$

$\left(53 - 2.131 \times \frac{\sqrt{10}}{\sqrt{16}}, 53 + 2.131 \times \frac{\sqrt{10}}{\sqrt{16}}\right) \Rightarrow (51.315, 54.6$

The heights of 10 males of a given locality are given by 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches is it reasonable to believe that the avg. height is greater than 64 inches. Test at 5% level of significance.

→ Null Hypothesis (H_0): The avg height of males is 64 inches.

Alternative Hypothesis (H_1): The avg height of males is greater than 64 inches. ($\mu > \mu_0$) (OTT)

Level of Significance (α): 5%.

Critical Value: CV of t (t_α) at 5% LOS with 9 degrees of freedom is 1.833.

Test Statistics: $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

$= \frac{1}{9} \times 90 = 10.$

$t = \frac{66 - 64}{\sqrt{10/10}} = 2.$

Conclusion: Reject H_0, since the absolute value is greater than the tabulated value.

Let $\bar{x}, \bar{y}$ be the two sample means of size n_1 & n_2 respectively which are drawn from the normal population having the mean of μ_1 & μ_2 and variance is σ_1^2 & σ_2^2.

→ Null Hypothesis: There is no significance difference of means. ($\mu_1 = \mu_2$)

Alternative Hypothesis: There is a significance difference of means.

$\mu_1 \neq \mu_2$ (TTT)

$\mu_1 > \mu_2$ (OTT)
$\mu_1 < \mu_2$

Level of Significance: If α not given, assume 5%.

Critical Value: t_α - OTT
$t_{\alpha/2}$ - TTT } at α level of significance with $\nu = n_1 + n_2 - 2$.

Test statistic: $t = \dfrac{\bar{x} - \bar{y}}{\sqrt{s^2(1/n_1 + 1/n_2)}}$

~~(OR)~~

where $s^2 = \frac{1}{n_1} + \frac{1}{n_2} - 2$
$\left[\sum (x_1 - \bar{x})^2 + \sum (y_1 - \bar{y})^2\right]$

(OR) $\dfrac{1}{n_1 + n_2 - 2} \left[(n_1 - 1) s_1^2 + (n_2 - 1) s_2^2\right]^2$

Conclusion: If calculated value of t is less than the tabulated value of t then accept H_0 else reject.

A sample of 2 electric light bulbs were tested for the length of life & the following data obtained:

Type I	Type II
$n_1 = 8$	$n_2 = 7$.
$\bar{x} = 1234$ hrs	$\bar{y} =$
$s_1 = 36$ hrs	$s_2 = 40$ hrs

Is the difference of means sufficient than ~~was~~ type I is superior than to type II regarding the length of life.

→ NH: The avg lifetime of both the Type I and Type II brands are same.

AH: Type I is superior than Type II. $(\mu_1 > \mu_2)$. (OTT).

LOS: $\alpha = 5\%$.

CV: The cv of t (t_α) at 5% LOS with 13 degrees of freedom is 1.771.

TS: $t = \dfrac{\bar{x} - \bar{y}}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ where $s^2 = \dfrac{1}{n_1 + n_2 - 2}\left[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2\right]$

$\boxed{s^2 = 1436.30}$

$t = \dfrac{1234 - 1036}{\sqrt{1436.30\left(\frac{1}{8} - \frac{1}{7}\right)}}$

$= 10.01$.

Conclusion: Reject H_0. $(10.01 > 1.771)$. Hence I is superior than II in length of life.

Two horses A & B were tested according to the time (seconds) to run on a particular track and the following results were obtained:

Horse A	28	30	32	33	33	29	39
Horse B	29	30	30	24	27	28	

Test whether the two horses have the same running capacity or not. $n_1 = 7$, $n_2 = 6$

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{s^2\left(\frac{1}{n_1} - \frac{1}{n_2}\right)}}$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	y_i	$y_i - 5$	$(y_i - \bar{y})^2$
28	-4				
30	-2				
32	0				
33	1				
33	1				
29	-3				
39	7				

→ NH: Both horse A & horse B have same running capacity.

AH: Horse A & Horse B do not same running capacity. $(\mu_1 \neq \mu_2)$ (TTT).

LOS: Assume $\alpha = 5\%$.

CV: The cv of t $(t_{\alpha/2})$ at 5% LOS with 11 degrees of freedom is 2.01.

TS: $t = \dfrac{\bar{x} - \bar{y}}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ where, $s^2 = \dfrac{1}{n_1 + n_2 - 2}$

$$\left[(x_i - \bar{x})^2 + \Sigma (y - \bar{y})^2\right] \text{—} ①$$

from ①;

$$t = \frac{32 - 25}{\sqrt{9.63\left(\frac{1}{7} + \frac{1}{8}\right)}} \Rightarrow 2.32$$

$$s^2 = \frac{1}{11}\left[80 + 26\right] = \frac{106}{11} = 9.63.$$

Conclusion: Accept H0.

A group of 5 patients treated with medicine A weights 42, 48, 39, 60, 41 and 2n group of 7 patients from same hospital treated with medicine B weights 38, 42, 56, 64, 68, 69, 62. Do you agree with the claim that medicine B increases the ~~weight~~ grade significantly.

→ :

Medicine A: 42 48 39 60 41

Medicine B: 38 42 56 64 68 69 62

$n_1 = 5$ $n_2 = 7$.

H0: Both medicines A & medicines B are same.

H1: Medicine A and medicine B is not same (TTT).

LOS: $\alpha = 5\%$.

CV = The
deg

TS: $t =$

Find the
of size
given tha
→ NH:

AH:

LOS:

CV:

TS:

CV = The cv of t ($t_{\alpha/2}$) at 5% LOS with 10 degrees of freedom is 1.85 n_1+n_2-2
$\Rightarrow 5+7-2 = 10.$

TS: $t = \dfrac{\bar{x}-\bar{y}}{\sqrt{s^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$

Find the max. difference of means of samples of size 10 & 12 in the probability 0.95 and given that the SD are found to be 2 & 3 respectively.

→ NH: The 2 samples are drawn from same normal population.

AH: The 2 samples are not drawn from same normal population (TTT)

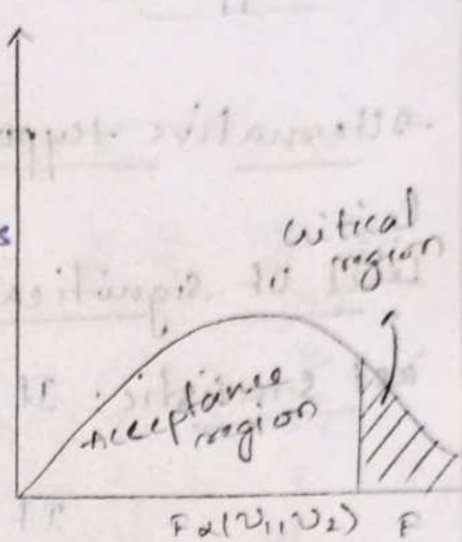
LOS: α = ~~0.95~~ 5%.

CV: The cv of t ($t_{\alpha/2}$) at 5% LOS with 20 DOF is 2.086.

TS: $t = \dfrac{\bar{x}-\bar{y}}{\sqrt{s^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$ ≥

F-Test / Variance Test:

Properties of F-Test:



→ F distribution is free from the population parameters & it is depend on the degrees of freedom

→ F distribution completely lies in 1st Quadrant only.

→ The F curve not only depends on the degrees of freedom but also all the order which they are stated.

$$F_{1-\alpha}(\nu_1,\nu_2) = \frac{1}{F_\alpha(\nu_2,\nu_1)}$$

→ The area under the curve is unity.

$$\int_0^\infty P(F)\,dF = 1.$$

→ The mode of F-distribution is less than unity.

Q) $F_{0.05}(7,5)$ dof $= 4.85$

Q) $F_{0.01}(24,18)$ dof

Q) $F_{0.95}(19,24)$ dof $= F_{1-0.05}(19,24) = \dfrac{1}{F_{0.05}(24,19)}$

$= \dfrac{1}{2.11} = 0.47.$

We can use the F-distribution to test of hypothesis population, variances same or not.

Null Hypothesis: Population variance are same ($\sigma_1^2 = \sigma_2^2$).

Alternative Hypothesis: Population variances are not same.

Level of significance: Assum $\alpha = 5\%$ (if not given)

Test Statistic: If $s_1^2 > s_2^2$; $F = \frac{s_1^2}{s_2^2}$

If $s_1^2 < s_2^2$; $F = \frac{s_2^2}{s_1^2}$

that is, F is equal to $\frac{\text{greater variance}}{\text{smaller variance}}$

Critical Value: Find critical value of F at α level of significance with (greater variance size − 1), (smaller variance size − 1).

Conclusion: If calculated value of F is less than tabulated value of F then accept H_0 else reject H_0.

2) The time taken by worker in performing a job by method I & method 2 given below:

M I	20	16	26	27	23	20	
M II	27	30	42	35	32	34	38

The data shows that the variance of time distribution is same or not

→ H_0: The 2 population variances are same. ($\sigma_1^2 = \sigma_2^2$)

H_1: The 2 population variances are not same. $(\sigma_1^2 \neq \sigma_2^2)$.

LOS: Since α = not given, assume $\alpha = 5\%$.

TS:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
20	-2	4	27	-7	49
16	-6	36	30	-4	16
26	4	16	42	8	64
27	5	25	35	1	1
23	1	1	32	-2	4
20	-2	4	34	0	0
			38	4	16

$\bar{x} = \frac{\sum x_i}{5} = 22$ $\quad \sum (x_i - \bar{x})^2 = 86$ $\quad \bar{y} = 34$ $\quad \sum (y_i - \bar{y})^2 = 150$

Now, $s_1^2 = \frac{1}{n_1 - 1} \sum (x_i - \bar{x})^2$

$= \frac{1}{5} \times 86 = 17.2$

$s_2^2 = \frac{1}{n_2 - 1} \sum (y_i - \bar{y})^2$

$= \frac{1}{6}(150) = 25$

$F = \frac{s_2^2}{s_1^2} = \frac{25}{17.2} = 1.45$

CV: The critical value of F at 5% LOS with

Conclusion: The calculated value of F is less than the tabulated value of F then Accept NH(H_0).

NOTE: To discuss whether the 2 samples are drawn from same normal population it is enough to verify the population variances are same & population means are same.

Q) 2 random samples are given below.

Sample No	size	Sample Mean	sum of sqrs of SD
1	10	15	90
2	12	14	100

Test the samples are drawn from same normal population

→

The nicotin contains in mg of a sample of tobacoo are found to be as follows:

Sample A   24   27   26   21   25

Sample B   27   30   28   31   22   36

then it is said that the 2 samples are drawn from same normal population or not.

→ case I: Verification of population means are same or not. (t-test).

1. NH: There is no significance difference of means. $(\mu_1 = \mu_0)$.

2. AH: There is significance difference b/w means. $(\mu_1 \neq \mu_0)$ (TTT)

3. LOS: Assume $\alpha = 5\%$.

4. CV: $(t_{\alpha/2})$ the critical value of t at 5% LOS with 9 dof is 2.262.

5. TS: (t) $\quad t = \dfrac{\bar{x} - \bar{y}}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

where $s^2 = \dfrac{1}{n_1 + n_2 - 2}\left[\Sigma(x_i - x)^2 - \Sigma(y_i - y)^2\right]$ — ①

x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	y	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$
24	0.6	0.36	27	2	4
27	2.4	5.76	30	+1	1
26	1.4	1.96	28	-1	1
21	-3.6	12.96	31	2	4
25	0.4	0.16	22	-7	49
			36	7	49

$\bar{x} = 24.6$ $\quad \Sigma(x_i - \bar{x})^2 = 21.2$

$\bar{y} = 29$ $\quad \Sigma(y_i - y)^2 = 108$

Case 2: verification of population variances are same or not (F-test)

1. NH: $H_0 = \sigma_1^2 = \sigma_2^2$

2. AH: $H_1 = \sigma_1^2 \neq \sigma_2^2$

3. LOS: $\alpha = 5\%$

4. TS: $F = \frac{G.V}{S.V}$

$$S_1^2 = \frac{1}{n_1 - 1} \Sigma (x_i - x)^2 = \frac{1}{4}(21.2) = 5.3$$

$$S_2^2 = \frac{1}{n_2 - 1} \Sigma (y_i - y)^2 = \frac{1}{5}(108) = 21.6$$

$$\therefore F = \frac{S_2^2}{S_1^2} = \frac{21.6}{5.3} = 4.07$$

5. CV: The critical value of F at 5% LOS with (5,4) dof = 6.26.

6. Conclusion: since the Fcalculated value is less than the Ftabulated value then accept the null hypothesis (H_0).

From above we can conclude that the population means & population variances are same. Hence the 2 samples are drawn from the same normal population.

4 coins were tossed 160 times & the following results are obtained:

No. of heads	0	1	2	3	4
observed frequency (O_F)	17	52	54	31	6

Is the coins are unbaised?

→ NH (H_0): The coins are unbiased.

AH (H_1): The coins are biased

LOS (α): Assume $\alpha = 5\%$.

CV: The CV of χ^2 at 5% LOS with 3 dof is 7.815

$(n-1) = 4-1 = 3$.

IS: $\chi^2 = \sum \frac{(O_F - E_F)^2}{E_F}$

Probability of Binomial Distribution:

$P(X = x) = {}^nC_x \, p^x q^{n-x}, \; (x = 0, 1, 2, 3, 4, \dots n)$

$= {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}, \; x = 0, 1, 2, 3, 4$

$P(X=0) = {}^4C_0 \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

$P(X=1) = {}^4C_1 \left(\frac{1}{2}\right)^4 = \frac{4}{16}$

$P(X=2) = {}^4C_2 \left(\frac{1}{2}\right)^4 = \frac{6}{16}$

$P(X=3) = {}^4C_3 \left(\frac{1}{2}\right)^4 = \frac{4}{16}$

$P(X=4) = {}^4C_4 \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

The expected frequency of zero head out of 160 trials: $160 \times \frac{1}{16} = 10$.

" one head out of 160 trials

" $160 \times \frac{4}{16} = 40$, $160 \times \frac{6}{16} = 60$, $160 \times \frac{4}{16} = 40$, $160 \times \frac{1}{16} = 10$.

No. of heads	O_f	E_f	$O_f - E_f$	$(O_f - E_f)^2$	$\frac{(O_f - E_f)^2}{E_f}$
0	17	10	7	49	4.9
1	52	40	12	144	3.6
2	51	60	-9	81	1.35
3	31	40	-9	81	2.02
4	6	10	-4	16	1.6

$$\sum \frac{(O_f - E_f)^2}{E_f} = 13.47.$$

Conclusion: Since the calculated value of φ^2 is greater than the tabulated value of φ^2. Reject NH (H_0). The coins are biased.

Fit a poisson distribution for the following data and test goodness of fit with 5% of LOS.

→ NH (H_0): There is no significance diff. of O_f and E_f.

AH (H_1): There is significance diff. b/w O_f and E_f.

LOS (α): $\alpha = 5\%$

CV: $n = 5$, $n - 2 = 3$. ⇒ 7.815

The CV of φ^2 at 5% LOS is 3 ⇒ 7.815.

TS: $\varphi^2 = \sum \frac{(O_f - E_f)^2}{E_f}$

Probability mass function of ~~B~~ Poisson Distribution

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad (x = 0, 1, 2, \ldots n)$$

where $\lambda = \frac{\sum f_i x_i}{\sum f_i}$

$$\therefore \Rightarrow \frac{0\times 419 + 1\times 352 + 2\times 154 + 3\times 56 + 4\times 19}{419 + 352 + 154 + 56 + 19} \Rightarrow 0.9$$

$$P(X=0) = \frac{e^{-0.9}\,0.9^{0}}{0!} = 0.406$$

$$P(X=1) = \frac{e^{-0.9}\,0.9^{1}}{1!} = 0.36$$

$$P(X=2) = \frac{e^{-0.9}\,0.9^{2}}{2!} = 0.16$$

$$P(X=3) = \frac{e^{-0.9}\,0.9^{3}}{3!} = 0.04$$

$$P(X=4) = \frac{e^{-0.9}\,0.9^{4}}{4!} = 0.011$$

expected frequency of '0' = $1000 \times P(X=0)$
= $1000 \times 0.406 = 406$.

" " '1' = $1000 \times P(X=1)$
= $1000 \times 0.3 = 360$

" " '2' = $1000 \times P(X=2)$
= $1000 \times 0.16 = 160$

" " '3' = $1000 \times P(X=3)$
= $1000 \times 0.04 = 40$

" " '4' = $1000 \times P(X=4)$
= $1000 \times 0.011 = 11$.

Conclusion: Reject NH (H_0).

chi-square Test for independence of attributes:

The given attributes are independant to each other or there is no significance difference of frequencies. (OR) $O_F = E_F$.

Alternative Hypothesis:

The given attributes are not independant to each other or there is significance difference b/w both frequencies. (OR) $O_F \neq E_F$

Level of Significance:

If α is not given, assume 5%.

critical value:

Find the critical value of φ^2 at α level of significance with $(R-1)(C-1)$ dof where $R \to$ No. of rows, $C \to$ No. of columns.

Test statistic: $\varphi^2 = \sum \frac{(O_F - E_F)^2}{E_F}$

Conclusion:

If the ~~calt~~ calculated value of φ^2 is less than the tabulated value of φ^2, accept NH(H_0) else reject NH(H_0).

NOTE: The E_F of each cell $= \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$

On the basis of information given below about the treatment of 200 patients suffering from a disease state whether the new treatment is comparitively to the conditional treatment.

	Favourable	Non favorable	Total
New	60	30	90
conventional	40	70	110
Total	100	100	200 grand Total

→ NH(H_0): Both new & conventional treatment independent of each other.

AH(H_1): New & conventional treatment are not independent to each other.

LOS(α): Assume 5%.

CV: The cv of φ^2 at 5% LOS with $(2-1)(2-1) = 1$ dof is 3.841

TS: $\varphi^2 = \sum \frac{(O_F - E_F)^2}{E_F}$

Expected frequencies of 60 = $\frac{90 \times 100}{200} = 45$

" " " 30 = $\frac{90 \times 100}{200} = 45$

" " " 40 = $\frac{110 \times 100}{200} = 55$

" " " 70 = $\frac{110 \times 100}{200} = 55$

O_F	E_F	$O_F - E_F$	$(O_F - E_F)^2$	$\frac{(O_F - E_F)^2}{E_F}$
60	45	15	225	5
30	45	-15	225	5
40	55	-15	225	4.09
70	55	15	225	4.09

$$\varphi^2 = \sum \frac{(O_F - E_F)^2}{E_F} = 18.18.$$

Conclusion: Since the calculated value is greater than tabulated value. Reject NH(H0). Both the treatments are dependent to each other.

4 methods are under development for making disks for super conducting material. 50 disks are made by each method & they are checked for superconductivity when cooled with liquid.

	Ist method	IInd method	IIIrd method	IVth method	Total
Success	31	42	22	25	120
failed	19	8	25	25	80
Total	50	50	50	50	200

Test the significance difference b/w the proportions of superconductors at .05 level.

→ NH(H0): There is no significance difference of proportions of superconductors ($P_1 = P_2 = P_3 = P_4$)

AH(H1): There is significance difference of proportions of superconductors. ($P_1 \neq P_2 \neq P_3 \neq P_4$)

LOS(α): 5%.

CV: The CV of φ^2 at 5% LOS with $(2-1)(4-1) = 3$ dof is 7.815

TS: $\varphi^2 = \sum \frac{(O_F - E_F)^2}{E_F}$

Expected frequencies of $31 = \frac{120 \times 50}{200} = 30$,

$42 = \frac{120 \times 80}{200} = 30$, $22 = \frac{120 \times 50}{200} = 20$, $25 = \frac{120 \times 50}{200} = 20$, $19 = \frac{80 \times 50}{200} = 20$

$8 = \frac{80 \times 50}{200} = 20$, $25 = \frac{80 \times 50}{200} = 20$, $25 : \frac{80 \times 50}{200} = 20$.

O_F	E_F	$O_F - E_F$	$(O_F - E_F)^2$	$\frac{(O_F - E_F)^2}{E_F}$
31	30	1	1	0.03
42	30	12	144	4.8
22	20	2	4	~~2.13~~ 4.8
25	20	5	25	~~0.05~~ 2.13
19	20	-1	1	~~7.2~~ 0.05
8	20	-12	144	~~3.2~~ 7.2
28	20	8	64 ~~25~~	3.2
25	20	5	25	1.25

$n-1$

$\mu_1 - \mu_2 = D_0$

$z \geq 3.1$

ND

False

TDB TP = 0

$\frac{\bar{x} - \mu}{s/\sqrt{n}}$

$z \leq 1.645$

0

$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

$H_a = \mu_1 = \mu_2$

chi-squared

Methods of studying co-relation:

There are 2 methods for co-relation

Graphical methods are scattered diagram & a simple graph.

Mathematical methods are:

Scattered Diagram: It is a chart applied by 42 tables to find whether any relationship b/w given variables. In this diagram x variables are ploted on horizontal axis & Y variables are ploted on vertical axis.

Simple Graph: The values of 2 variables plotted on the graph paper we get 2 curves. One is for x variable & another is for Y variable. These 2 curves reveals the direction & closeness of the two variables and also reveal whether the variables are related or not. If both curves move in the same direction either upward / downward then the co-relation is positive. If they move in the opposite direction then the co-relation is negative.

Co-relation Coeffeicent

- A statical measure which is used for analyzing the behavior of 2 or more than 2 variables.

It is denoted by r.

Properties of Co-relation Coefficient:

→ It is denoted by r.

→ It lies b/w -1 to 1, $r \in [-1, 1]$.

→ If $r = 1$, then the co-relation is perfect in the positive direction.

→ If $r = -1$, then the co-relation is perfect in the negative direction.

→ If $r = 0$, then there is no co-relation b/w the given variables. In this case the variables are called independant variables.

→ If r lies $0 < r < 1$, $-1 < r < 1$, $-1 < r < 0$

then it is partial.

Karl Pearson's Co-relation Coefficient:

It is defined as
$$\frac{N\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[N\Sigma x^2 - (\Sigma x)^2][N\Sigma y^2 - (\Sigma y)^2]}}$$

Direct Method　　　　Assumed Method.

$$r = \frac{\Sigma xy}{\sqrt{(\Sigma x^2)(\Sigma y^2)}}$$

$\bar{x}, \bar{y} \in I$ (or) $\approx$

$\Sigma x = 0, \Sigma y = 0$

$N \rightarrow$ no: of. observations.
$X \rightarrow x_i - \bar{x}$
$Y \rightarrow y_i - \bar{y}$

Q) Find the co-relation coefficient for the following data:

x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	x^2	y^2	xy
100	98	+1	3	1	9	3
101	99	2	4	4	16	8
102	99	3	4	9	16	12
102	97	3	2	9	4	6
100	95	1	0	1	0	0
99	92	0	-3	0	9	0
97	95	-2	0	4	0	0
98	94	-1	-1	1	1	1
96	90	-3	-5	9	25	15
95	91	-4	-4	16	16	16

$\bar{x} = \frac{\Sigma x_i}{10} = 99.$ $\bar{y} = \frac{\Sigma y_i}{10} = 95.$

Since the means of x & y are integers so we can apply Karl Pearson's co-relation coefficient of direct method.

$$r = \frac{\Sigma xy}{\sqrt{(\Sigma x^2)(\Sigma y^2)}} = \frac{61}{\sqrt{54 \times 96}} = 0.84.$$

$X = x_i - \bar{x}$

$Y = y_i - \bar{y}$

$(-1 < r < 1)$ positive direction.

8) The following table gives the ~~sports~~ scores obtained by 11 students of maths, english

Score in English	Score in Maths	$x_i-\bar{x}$	$y_i-\bar{y}$	x^2	y^2	xy
40	45	-32	-10	1024	100	320
46	45	-36	-10	676	100	260
54	50	-18	-5	324	25	90
66	43	-12	-12	144	144	144
70	40	-2	-15	4	225	30
80	75	8	20	64	400	160
80	55	10	0	100	0	0
82	72	13	17	169	289	221
85	65	13	10	169	100	130
90	42	18	-13	324	169	-234
95	70	23	15	529	225	345

$\bar{x} = 71.5$ $\quad \bar{y} = 54.7$

The means of x & y are 71.5 & 54.7 respectively.

We apply Assumed method of Karl Pearson co-relation coefficient.

Let us assume that $\bar{x}$ is 71 or 72, and $\bar{y}$ is 55.

$x = x_i - \bar{x}$

$y = y_i - \bar{y}$

$$r = \frac{N\sum xy - (\sum x)(\sum y)}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$$

$$r = \frac{11 \times 1466 - (-5)(-3)}{\sqrt{(11 \times 3527 - 25)(11 \times 1771 - 17)}}$$

$= 0.58.$

Spearman's Rank Co-relation Coefficient:

The rank co-relation coefficient is denoted by ρ and it is defined as,

for non-repeated ranks = $\rho = 1 - \frac{6}{N(N^2-1)} \Sigma D^2$

for repeated ranks = $\rho = 1 - \frac{6}{N(N^2-1)} \left[\Sigma D^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2}{12}(m_2^2-1) + \dots \right]$

where, $N \rightarrow$ represents no. of observations

$D \rightarrow$ difference of ranks.

$m_i \rightarrow$ no. of tied ranks.

Properties of Rank Co-relation:

- → It is denoted by ρ.
- → It lies between $(-1, 1)$.
- → If $\rho = -1$, then there is a complete disagreement in the order of the ranks & in opposite direction.
- → If $\rho = 1$, then there is a complete agreement in the order of the ranks & in same direction.

Q) A random sample of 5 college students is selected and their grades in maths & statistics are found to be:

Maths	(R)	statistics	(R)	D $(R_1 - R_2)$	D^2
85	2	93	1	1	1
60	4	75	3	1	1
73	3	65	4	-1	1
40	5	50	5	0	0
90	1	80	2	-1	1

$$\rho = 1 - \frac{6}{N(N^2-1)} \Sigma D^2$$

$$= 1 - \frac{6}{5(5^2-1)} (4) = \frac{4}{5} = 0.8.$$

2) find the rank co-relation coefficient for the following data

x	(R)	y	(R)	D $(R_1 - R_2)$	D^2
65	9	68	5.5	3.5	12.25
63	11	66	9.5	1.5	2.25
67	6.5	68	5.5	1	1
64	10	65	11.5	-1.5	2.25
68	4.5	67	3	1.5	2.25
62	12	66	9.5	2.5	6.25
70	2	68	5.5	-3.5	12.25
66	8	65	11.5	3.5	12.25
68	4.5	71	1	-2.5	6.25
67	6.5	67	8	-2.5	6.25
69	3	68	5.5	-1.5	12.25
71	1	70	2	-1	1

In the given data of X-series 68 is repeated twice & also 67 is repeated twice and in Y-series 66 & 65 are repeated twice & 68 is repeated 4 times.

X	Y
68 - 2	66 - 2
67 - 2	65 - 2
	68 - 4

$$\rho = 1 - \frac{6}{N(N^2-1)}\left[\Sigma D^2 + \frac{m_1}{12}(m_1^2-1) + \frac{m_2}{12}(m_2^2-1) + \frac{m_3}{12}(m_3^2-1) + \frac{m_4}{12}(m_4^2-1) + \frac{m_5}{12}(m_5^2-1)\right]$$

$m_1 = m_2 = m_3 = m_4 = 2$.

$m_5 = 4$.

$$\rho = 1 - \frac{6}{12(12^2-1)}\left[\Sigma D^2 + \frac{4(2)}{12}(2^2-1) + \frac{4}{12}(4^2-1)\right]$$

$$1 - \frac{6}{12(12^2-1)}\left[\Sigma D^2 + \frac{8}{12}(3) + \frac{4}{12}(15)\right]$$

$$1 - \frac{6}{12(12^2-1)}\left[\Sigma 72.5 + \frac{8}{12}(3) + \frac{4}{12}(15)\right]$$

$$\Rightarrow 1 - \frac{6}{1716}\left[\Sigma 72.5 + 2 + 5\right]$$

$$\Rightarrow 1 - 0.0034\,[79.5] \Rightarrow 0.729.$$

UNIT - VI Curve Fitting And Regression:

The process or the methods for finding the unknown curve that satisfies given set of points is known as curve fitting

Fitting a Straight Line:

Let us assume that the required straight line of the form by $y = a + bx$ where a & b are arbitary constants.

By the method of least squares the normal equations of straight line are given by,

$\Sigma y = na + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

Solve the above normal equations for two unknowns (a, b) which gives the required straight line.

Similarly, one of the normal equations of $ax + b$:

$\Sigma y = a\Sigma x + nb$

$\Sigma xy = a\Sigma x^2 + b\Sigma x$

· Find the straight line which fits the given data:

x	10	12	13	16	17	20	25
y	10	22	24	27	29	33	37

→ Let us assume that the required straight line of the form $y = ax + b$

By the method of least squares the normal equations have straight line is given by:

$$\left.\begin{aligned}\Sigma y &= a\Sigma x + nb \\ \Sigma xy &= a\Sigma x^2 + b\Sigma x\end{aligned}\right\} \text{——} (1).$$

x	y	x^2	xy
10	10	100	100
12	22	144	264
13	24	169	312
16	27	256	432
17	29	289	493
20	33	400	660
25	37	600	925
$\Sigma x =$ 113	$\Sigma y =$ 182	$\Sigma x^2 =$ 19803	$\Sigma xy =$ 3186

from (1):

$$182 = 113a + 7b \quad \times 113$$

$$3186 = 19803a + 113b \quad \times 7$$

$a \Rightarrow 1.56$

$b \Rightarrow 0.79$

$\therefore$ The required straight line is $y = 1.56x + 0.79$.

find a straight line $y = a + bx$ for the following data:

x	78	77	85	88	87	82	81	77	76	83	97	93
y	84	82	82	85	89	90	88	92	83	89	98	99

$\rightarrow$ $y = a + bx$

$\Sigma y = na + b\Sigma x$, $\Sigma xy = a\Sigma x + b\Sigma x^2$

x	y	x^2	xy
78	84	6084	6552
77	82	5929	6314
85	82	7225	6970
88	85	7744	7480
87	89	7569	7743
82	90	6724	7380
81	88	6561	7128
77	92	5929	7084
76	83	5776	6308
83	89	6889	7387
97	98	9409	9506
93	99	8649	9207
$\Sigma x = 1004$	$\Sigma y = 1061$	$\Sigma x^2 = 84{,}488$	Σxy^2 89,054

from ①:

$$1061 = 12a + 1004b$$

$$89{,}059 = 1004a + 84{,}488b$$

$$a = 38.789$$

$$b = 0.593$$

$$\therefore y = a + bx$$

$$y = 38.789 + 0.593x.$$

Fitting A Parabola:

Let us assume that the second degree polynomial of the form $y = a + bx + cx^2$ (a, b, c)

By the method of least squares the normal equations of parabola

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^2 + c\Sigma x^4$$

Solve the above 3 equations for 3 unknowns a, b, c. It is the required parabola.

Similarly the normal equations of parabola of the form $y = ax^2 + bx + c$

$$\Sigma y = a\Sigma x^2 + b\Sigma x + nc$$

$$\Sigma xy = a\Sigma x^2 + b\Sigma x^2 + c\Sigma x$$

$$\Sigma x^2 y = a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2$$

Solving the above equations we get the required parabola.

Q) Fit the 2nd degree polynomial for the following data

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

Let us assume that the required 2nd degree polynomial of the form $y = a + bx + cx^2$.

By the method of least squares the normal equations of parabola are given by:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^2 + c\Sigma x^4$$

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	6.3	16	64	256	25.2	100.8

$\Sigma x = 10$, $\Sigma y = 12.9$, $\Sigma x^2 = 30$, $\Sigma x^3 = 100$, $\Sigma x^4 = 354$,
$\Sigma xy = 37.1$, $\Sigma x^2 y = 130.3$

from (i):

$$12.9 = 5a + 10b + 30c$$
$$37.1 = 10a + 30b + 100c$$
$$130.3 = 30a + 100b + 354c$$

$a = 1.42$
$b = -1.07$
$c = 0.55$

$$y = 1.42 + (-1.07)x + (0.55)x^2$$

Find the 2^{nd} degree polynomial of the following data:

x	1	2	3	4	5
y	10	12	8	10	14

→

x	y	x^2	x^3	x^4	xy	x^2y
1	10	1	1	1	10	10
2	12	4	8	16	24	48
3	8	9	27	81	24	72
4	10	16	64	256	40	160
5	14	25	125	625	70	350

$\Sigma x = 15$, $\Sigma y = 54$, $\Sigma x^2 = 55$, $\Sigma x^3 = 225$, $\Sigma x^4 = 979$,
$\Sigma xy = 168$, $\Sigma x^2 y = 640$.

from (i):

$54 = 5a + 15b + 55c$

$168 = 15a + 55b + 225c$

$640 = 55a + 225b + 979c$

$a = 14, \ b = -3.68, \ c = 0.71$.

$y = 14 + (-3.68)x + (0.71)x^2$.

Regression:

In the correlation we have discussed the strength of variables & the direction of the variables. In regression we can discuss the value of the dependent variable where the independent variable is known.

The process of finding the value of the dependent variable if the value of the independent variable is known as Regression.

We have two types of regression lines.

Regression line on Y on X

Regression line on X on Y.

Regression Lines:

→ The regression line of y on x is given by

$$Y - \bar{Y} = r \frac{\sigma_Y}{\sigma_x} (X - \bar{X}) \quad \text{(OR)}$$

$$Y - \bar{Y} = b_{yx} (X - \bar{X})$$

where, $X = x_i - \bar{x}$, $Y = y_i - \bar{y}$, r is correlation cofficient σ_x & σ_y are SD of x & y series, and b_{yx} is regression cofficienat of Y on X.

$$b_{yx} = \frac{N \Sigma XY - (\Sigma X)(\Sigma Y)}{N \Sigma X^2 - (\Sigma X)^2}$$

→ The regression line of x on y is given by, $X - \bar{X} = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y})$ (OR)

$$X - \bar{X} = b_{xy} (Y - \bar{Y})$$

$$b_{xy} = \frac{N \Sigma XY - (\Sigma X)(\Sigma Y)}{N \Sigma Y^2 - (\Sigma Y)^2}$$

$r = \sqrt{b_{yx} \times b_{xy}}$

Q) Given the following information. find the regression equations of x & y. and also given that the correlation coefficient 0.9. The means of statistics & economics are 40 & 8. The SD of marks of statistics & economics are 12 & 16.

→ Let us assume that the marks of statistics and economics are x & y.

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 40 = 0.9 \frac{12}{16} (y-8)$$

$$x - 40 = 0.675 (y-8)$$

$$x = 0.675y - 8(0.675) + 40$$

$$x = 0.675y + 34.6.$$

Q) The following data related to the marks of 10 students in internal test & university examination for the maximum of 50 each.

IM:	25	28	30	32	35	36	38	39	42	45
UM:	20	26	29	30	25	18	26	35	~~46~~ 35	46

Find the two regression lines & determine

i) The most likely I.M for University ~~month~~ mark of 25

ii) The " " U.M for internal mark of 30.

⇒ Let us assume that x denotes IM. & y denotes UM of a student.

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	XY	X^2	Y^2
25	20	-10	-9	90	100	81
28	26	-7	-3	21	49	9
30	29	-5	0	0	25	0
32	30	-3	+1	3	9	1
35	25	0	-4	0	0	16
36	18	1	-11	-11	1	121
38	26	3	-3	-9	9	9
39	35	4	6	24	16	36
42	35	7	6	42	49	36
45	46	10	17	170	100	289
$\bar{x} = 35$	$\bar{y} = 29$	$\Sigma X = 0$	$\Sigma Y = 0$	$\Sigma XY = 330$	$\Sigma X^2 = 358$	$\Sigma Y^2 = 598$

The regression line of y on x

$$x - \bar{y} = b_{yx}(x - \bar{x})$$

where $b_{yx} = \dfrac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{N\Sigma X^2 - (\Sigma X)^2}$

$$= \frac{10 \times 330 - (0)(0)}{10 \times 358} = 0.92$$

$$Y - \bar{Y} = 0.92(X - \bar{X})$$

$$Y - 0 = 0.92(X - 0)$$

$$Y = 0.92X.$$

$$y - \bar{y} = 0.92(x - \bar{x})$$

$$y - 29 = 0.92(x - 35)$$

$$y = 0.92x - 0.92 \times 39 + 29$$

$$y = 0.92x - 3.2$$

$\therefore$ If $x = 30$ then, $y = 0.92 \times 30 - 3.2 = \underline{24.4}$.

∴ The UM & IM of 30 is 24.4

The regression line of x on y:

$$x-\bar{x} \quad b_{xy}(y-\bar{y})$$

$$b_{xy} = \frac{N\Sigma xy-(\Sigma x)(\Sigma y)}{N\Sigma y^2-(\Sigma y)^2}$$

$$= \frac{10 \times 324}{10 \times 598} = 0.54$$

$$x-\bar{x} = 0.54(y-\bar{y})$$

$$x-0 = 0.54(y-0)$$

$$x = 0.54\cancel{x}$$

$$x-\bar{x} = 0.54(y-\bar{y})$$

$$x-35 = 0.54(y-29)$$

$$x = 0.54y - 29(0.54) + 35$$

$$\therefore x = 0.54y + 19.34$$

If $y = 25$, then $x = 0.54 \times 25 + 19.34 = 32.84$

Let us assume that x denotes IM & y denotes UM of student then the regression line of UM on IM is given by

$$y = ax + b$$